Machine Learning Project

Title: Automatic Image Colorization via Chrominance Discretization and Color Prior Re-weighting in U-Net Architectures.

Abstract:

Image colorization is a complex, ill-posed problem due to the high degree of uncertainty in mapping luminance to chrominance. This project presents an automatic image colorization system utilizing a U-Net–style convolutional neural network optimized for the CIE LAB color space. Unlike standard regression models that often yield desaturated results, this approach treats colorization as a multinomial classification task by discretizing the LAB color channels into distinct bins. To address the inherent class imbalance in natural colors, a color-prior reweighting scheme is implemented during training. The system was trained on a diverse dataset and further enhanced through post-processing techniques. Results demonstrate that the proposed model produces more vibrant and semantically plausible colorizations than traditional regression-based methods, thereby effectively capturing the diversity of natural scenes.

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1 Introduction

Image colorization involves mapping a single-channel grayscale image to a multi-channel color output. Because many different colors can share the same brightness level, the problem is **mathematically illposed**. While traditional methods required human intervention, modern deep learning allows us to learn "color logic" from massive datasets.

In this project, we implement an automated colorization pipeline based on a **U-Net architecture**. Our approach moves away from traditional color regression in favor of a classification-based strategy. The core contributions of this project include:

- **LAB Space Processing:** Utilizing the CIE LAB color space to isolate luminance and simplify the learning objective.
- **Chrominance Discretization:** Formulating color prediction as a classification problem by binning the a and b channels, allowing the model to handle multi-modal color distributions.
- **Class Re-balancing:** Implementing a color prior re-weighting scheme to prevent the model from defaulting to "safe," desaturated colors.
- **Refined Output:** Utilizing post-processing techniques to ensure the final images are both vibrant and spatially consistent.

2. Related Work

2.1 Traditional Manual and Reference-Based Methods

Early image colorization methods relied heavily on manual user intervention, such as "scribbling" colors onto specific regions, or on handcrafted rules to propagate color across pixels with similar intensities. While these methods could produce high-quality results, they were not scalable; the output quality was directly tied to the effort and artistic skill of the human user.

2.2 Regression-Based Deep Learning

With the advent of Deep Learning, researchers moved toward fully automatic systems. Initial approaches utilized Convolutional Neural Networks (CNNs) to treat colorization as a regression task, attempting to predict exact, continuous values for the chrominance channels. However, these models suffered from the "mean-averaging" problem: when the model was unsure whether a car should be red or blue, it would predict an average of the two (Gray), resulting in dull, desaturated images.

2.3 Classification and Color Discretization

To overcome the limitations of regression, modern frameworks—most notably Zhang et al. (2016)—reformulated colorization as a multinomial classification problem. By discretizing the ab color space into distinct bins, the model can predict a probability distribution over multiple possible colors. This allows for "multimodal" results—recognizing that an object could be one of several vibrant colors rather than a single gray average. This approach also introduced color prior reweighting, which forces the model to learn rare, vibrant colors rather than overrepresented neutral tones.

2.4 Encoder-Decoder Architectures (U-Net)

For dense pixel-level prediction, the U-Net architecture has become a standard. Its use of skip connections allows the network to preserve high-resolution spatial details from the input while simultaneously learning high-level semantic context (e.g., "this shape is a tree, so it should be green").

2.5 Proposed Approach

The present work builds upon these milestones by integrating the probabilistic classification logic of Zhang et al. into a U-Net–style architecture. This combination aims to leverage the classification approach's vibrant color diversity while maintaining the superior spatial resolution and boundary consistency provided by skip connections.

3. Methodology & System Design

3.1 Color Space Representation

The choice of color space plays a critical role in image colorization. In this work, the *CIE LAB* color space is adopted instead of the standard RGB representation due to its perceptual properties and its effectiveness in separating luminance from chrominance.

3.1.1 Limitations of RGB for Colorization

In the RGB color space, color information is represented using three highly correlated channels (RGB) that jointly encode both brightness and chromatic content. This coupling makes it difficult to isolate structural information from color information, complicating the learning process. Additionally, RGB is not perceptually uniform; small mathematical changes in RGB values do not correspond linearly to how humans perceive color differences. Consequently, directly predicting RGB values often leads to unstable training and visually inconsistent results.

3.1.2 Lab Color Space

The LAB color space represents color using three distinct components:

- L (Lightness): Captures brightness and structural information (the grayscale image).
- A (Chrominance): Represents the Green–Red axis.
- B (Chrominance): Represents the Blue–Yellow axis.

A key property of this space is the explicit decoupling of luminance from chrominance. This allows the colorization model to use the L channel as a structural guide while focusing its predictive power exclusively on the A and B channels.

3.1.3 Technical Advantages

Using the LAB color space provides several advantages for our classification-based approach:

1. Decoupled Learning: The luminance channel contains the edges and textures needed to identify objects without interference from chromatic data.

2. Perceptual Uniformity: Distances in Lab space closely reflect human perception, ensuring that the model's loss function aligns with visual quality.
3. Simplified Objective: Predicting only two channels (A, B) instead of three (RGB) reduces the dimensionality of the output space, making the discretization (binning) process more efficient.

3.1.4 Integration with the Proposed Model

During the preprocessing phase, input images are converted from RGB to LAB. Only the normalized L channel is provided to the U-Net as input. The model then predicts a probability distribution over the discretized A,B bins. In the final Inference stage, the predicted chrominance values are combined with the original high-resolution L channel and converted back to RGB for final visualization.

3.2 Problem Formulation

The objective of this project is to automatically colorize grayscale images by predicting plausible chrominance information based solely on luminance input. Given an image represented in the CIE LAB color space, the task is to infer the chroma components a and b from the luminance channel L.

3.2.1 Input Representation

Let $L \in \mathbb{R}^{H \times W}$ denote the normalized luminance channel of an image, where H and W are the image height and width. The model input is a single-channel grayscale image obtained by converting an RGB image to the LAB color space and scaling the luminance channel to the [0,1] range.

By isolating luminance from chrominance, the model is required to learn semantic color relationships rather than brightness information.

3.2.2 Output Representation and Color Quantization

Instead of directly regressing continuous chroma values, the chrominance space is discretized into a finite number of bins. The A and B channels are each quantized into N uniform bins, resulting in $K = N^2$ discrete color classes.

Each pixel’s chroma value is mapped to a class index using:

$$c = A_{bin} \times N + B_{bin}$$

This formulation transforms the colorization task into a pixel-wise multi-class classification problem, thereby enabling the model to represent uncertainty and multimodal color distributions.

3.2.3 Model Prediction

The model is defined as a function:

$$f_{\theta}: L \to P$$

where $P \in \mathbb{R}^{H \times W \times K}$ represents a probability distribution over the color classes for each pixel. A U-Net–based convolutional neural network is used to learn this mapping. The encoder extracts hierarchical semantic features from the luminance input, while the decoder reconstructs spatial resolution using skip connections to preserve fine-grained details. The final layer produces K logits per pixel, which are converted to class probabilities using the softmax function.

3.2.4 Loss Function and Color Prior Re-weighting

Training is performed using a pixel-wise cross-entropy loss. However, natural images exhibit a highly imbalanced color distribution, with desaturated colors appearing significantly more frequently than saturated ones.

To address this imbalance, a color prior distribution is computed over the training dataset by accumulating a histogram of quantized chroma values. Let p_k denote the empirical probability of color class k. The loss is re-weighted using:

$$w_k = \frac{1}{(p_k^{\alpha} + \epsilon)}$$

The weighted cross-entropy loss is defined as:

$$\mathcal{L} = -\sum_{k=1}^{K} w_k \; y_k \log(p_k)$$

This encourages the model to predict rare yet visually salient colors, thereby improving overall color diversity and realism.

3.2.5 Inference and Color Reconstruction

During inference, the model outputs a probability distribution over color classes for each pixel. Instead of selecting the most likely class, a soft expectation is computed over the chroma bin centers:

$$\hat{A}, \hat{B} = \sum_{k=1}^{K} p_k \cdot (A_k, B_k)$$

Temperature scaling is applied to the softmax output to control color confidence. The predicted chroma maps are further refined using edge-aware bilateral filtering to reduce noise while preserving boundaries. To enhance color vividness, a chroma gain factor is applied, followed by clipping to prevent color artifacts. The final colorized image is reconstructed by combining the predicted a and b channels with the original luminance channel and converting the result back to the RGB color space.

3.2.6 Learning Objective

The overall objective is to learn model parameters θthat minimize the expected weighted classification loss over the training dataset: $\theta^* = \arg\min_{\theta} \mathbb{E}_{(L,c)}[\mathcal{L}(f_\theta(L), c)]$

This unified formulation captures the full colorization pipeline from grayscale input to final color reconstruction.

4. Implementation Details

This section describes the practical implementation of the project, including the curation of the training data, the automated preprocessing pipeline, and the details of data preparation for the U-Net model.

4.1 Dataset Composition

To ensure robust and semantically meaningful color prediction, the model was trained on a large-scale and diverse image dataset covering a wide range of scenes, objects, and color distributions. The dataset was carefully curated and preprocessed to maximize color quality and minimize noise that could negatively affect training.

Data Sources: The training corpus was assembled from five complementary sources:

- **COCO 2017**: The primary source, providing complex scenes with multiple interacting objects.
- **Flickr30k**: Included to provide diverse real-world imagery, particularly outdoor scenes and humancentered activities.
- **Flickr Faces HQ (FFHQ)**: Integrated specifically to improve the model's accuracy in predicting realistic skin tones.
- **102 Flowers Dataset**: Added to introduce highly saturated and fine-grained color examples, improving performance on botanical and natural scenes.
- **Custom Internet Image Collection**: Additional images were collected via public APIs to address underrepresented color distributions and scene types within the existing datasets.

This multi-source strategy helped reduce dataset bias and improved the model's robustness across different visual domains.

4.2 Automated Dataset Cleaning Pipeline

Raw image datasets often contain unsuitable samples that can degrade model performance. To address this, a custom Python-based automated cleaning pipeline was developed to filter low-quality images while preserving valid data for training.

1. Resolution Filtering: The model is trained using a fixed input resolution of 256 × 256 pixels. This step ensures consistent spatial quality across all training samples.

- The script recursively scans the dataset directory for image files (.jpg, .jpeg, .png).
- Image dimensions are extracted using OpenCV.
- Images with a width or height smaller than 256 pixels are excluded, because they require aggressive upscaling, which introduces artifacts and reduces color fidelity.
- Instead of deletion, rejected images are moved to a dedicated removed/ directory to allow manual inspection if needed.

2. Grayscale Image Removal

Since the training objective is to learn the mapping from grayscale to color, the dataset must provide valid color supervision.

- Images that are already grayscale or heavily desaturated are removed.
- This prevents the model from being "tricked" into learning trivial mappings, which often results in the "sepia-tone" or washed-out outputs common in uncleaned datasets.
- Removing grayscale images significantly reduces the risk of producing washed-out or sepia-toned colorizations during inference.

4.3 Data Normalization and Color Binning

Once the dataset is cleaned, images undergo a standard transformation pipeline during the loading phase:

- Resizing: All images are resized to 256 × 256 using bicubic interpolation to preserve spatial detail.
- Color Space Conversion: Images are converted from RGB to the CIE LAB color space, which separates luminance from chrominance.
- Normalization: The luminance channel $L$is normalized to the range [0,1], to stabilize the gradient descent.
- Target Generation: The chrominance channels A and B are discretized into uniform bins of a predefined size. Each pixel is assigned a class index representing a specific (a, b) bin combination.

  The resulting class map serves as the ground-truth target for training, allowing the model to treat color prediction as a pixel-wise classification problem.

4.4 Color Space Quantization

To convert the continuous LAB color space into a discrete set of color classes, the a and b channels are quantized into N bins. This allows the model to treat colorization as a classification problem, rather than predicting continuous values for each pixel.

- Binning Process: The range of each channel (from -128 to 127 for both $a$and b) is divided into N equal bins. For example, with $N = 16$, there are 256 possible color classes (16×16).
- Class Mapping: Each pixel's chroma values are mapped to the nearest bin index:

$$\text{class} = a_{\text{bin}} \times N + b_{\text{bin}}$$

- Training Objective: During training, the model learns to predict a probability distribution over these classes for every pixel. This allows the model to capture the "multi-modal" nature of color (the fact that a shirt could be equally likely to be blue or red).

4.5 Training and Fine-Tuning Strategy

1- Training Strategy

The model was initially trained with a learning rate of 0.003 using the Adam optimizer. The training process followed a typical deep learning setup, including a learning rate scheduler to adjust the learning rate during training.

- Learning Rate Adjustment:
  The learning rate was initialized at 0.003 and manually adjusted at specified points during training. In particular, the Exponential Moving Average (EMA) (used for stabilizing training) was employed, which helped the model converge more smoothly. When the EMA was applied, the learning rate was lowered slightly to facilitate finer tuning as the model approached convergence. This dynamic adjustment helps prevent the model from overshooting the optimal solution, promoting better generalization.
- Training Process:
  The model was trained using the standard training loop, with cross-entropy loss and color-prior reweighting. The optimizer was set to Adam with the standard weight decay to prevent overfitting. A batch size of 20 was used, and training was carried out for 139 epochs, with the model periodically saved as checkpoints for resuming training if necessary.

1- Fine-Tuning Strategy

Once the base model was trained on the initial dataset, fine-tuning was performed to improve performance further. The fine-tuning process involved two key components:

2.1 Training on a Mixed Dataset:

The model was fine-tuned on a mix of: o Already-seen images: These images were part of the original training dataset, allowing the model to refine its predictions based on prior knowledge.

- New unseen images: This introduced diversity to the model and allowed it to adapt to new, previously unseen color distributions and lighting scenarios. By combining the two datasets, the model could generalize better to a wider variety of real-world images.

2.2 Using the Same Color Prior:

During fine-tuning, the model continued to use the same color prior used in the original training phase. This color prior, computed from the dataset, was used to address class imbalance by reweighting the cross-entropy loss function. The color prior helps the model focus more on underrepresented colors, ensuring that rare colors are learned more effectively.

The fine-tuning process was carried out with a lower learning rate (manually set during training) to avoid catastrophic forgetting, allowing the model to refine its learned features without losing the previously learned knowledge. The Exponential Moving Average (EMA) continued to be applied during fine-tuning to stabilize the model further and help improve its performance on the fine-tuning dataset.

2- Key Takeaways from the Strategy

- Training with a dynamic learning rate helped the model converge more efficiently, aided by EMA for smoother updates.
- Fine-tuning on mixed datasets with already seen and new images allowed the model to better generalize to new scenarios while refining its performance on familiar data.
- The use of the same color prior during both training and fine-tuning ensured consistent color reweighting, leading to more balanced and diverse color predictions.

5 Results and Evaluation

5.1 Limitations and Failure Cases

While the model performs well in many scenarios, there are still some limitations that need to be addressed:

1. Lighting Conditions:
    - The model struggles with inconsistent or poor lighting, leading to unrealistic color predictions. Areas with complex shadows or unusual light sources tend to have overconfident color assignments, producing unrealistic color blobs.
2. Ambiguous Objects:
    - Objects with similar luminance values but different expected colors (e.g., different shades of gray) can cause the model to produce ambiguous or incorrect colors.
3. Rare Color Prediction:
    - Despite the color prior re-weighting, rare or unique colors might still be underrepresented, resulting in less accurate predictions for uncommon objects or scenes.
4. Overfitting with Small Datasets:
    - The model's performance could be further improved with a larger, more diverse dataset. Training on a more diverse set of images would help reduce overfitting and improve generalization across different color distributions and lighting conditions.

5.2 Inference and post-processing

1- Model Inference

During inference, the trained model takes a grayscale image (Luminance channel L) as input and predicts the chrominance channels $a$and b in the LAB color space. The grayscale image is first pre-processed (resized and normalized) to match the input requirements of the trained model. Once the image is prepared, the model generates a probability distribution over a set of discrete color bins for each pixel. These predictions are then used to estimate the final chroma values (for both A and B channels) using a soft expectation approach, in which the bin centers weigh the predicted probabilities to compute the expected color values.

2- Post-Processing

Once the model outputs the predicted color values, several post-processing steps are applied to refine the results and improve the visual quality of the colorized image:

1. Bilateral Filtering:
   Bilateral filtering is applied to the predicted chroma values to reduce noise while preserving salient edges in the image. This process helps mitigate unwanted artifacts, such as sharp color transitions, thereby ensuring that the colorization appears more natural and consistent, particularly in fine-detail regions.
2. Chroma Gain Adjustment:
   To enhance the vibrancy and intensity of the predicted colors, a chroma gain factor is applied to the $a$and b channels. This adjustment amplifies the chrominance values, making the colorization appear more vivid and visually appealing. The chroma gain helps make the colorization output brighter and more realistic, particularly for objects with highly saturated colors.
3. Clipping:
   After the chroma gain is applied, the A and B values are clipped to ensure they remain within the valid range of the LAB color space. For example, the A and B values are constrained to the range $[-128,127]$, preventing any out-of-range color values that could distort the final output.
4. Upscaling:
   Since the model operates on resized images (e.g., 256x256), the predicted chroma channels are upscaled back to the original image size. This ensures that the colorized output matches the spatial resolution of the original input image, preserving image details while applying the chrominance values.
5. Reconstruction:
   The final step involves reconstructing the LAB image by combining the original Luminance channel L (which remains unchanged) with the upscaled A and B channels. The LAB image is then converted back into the RGB color space using standard color transformation algorithms. The resulting RGB image is the final colorized output.

Through these post-processing steps, the model's raw predictions are refined, leading to a more polished, vibrant, and realistic colorized image.

5.3 Qualitative Evaluation and Visual Results

The model was evaluated on a variety of grayscale input images, and the results show promising colorization quality. The following observations were made:

1. Object Color Recognition:

The model performs well in recognizing and colorizing objects within an image, with minimal to no color bleeding between adjacent objects. Object boundaries are clearly preserved, improving color separation between elements in the scene.

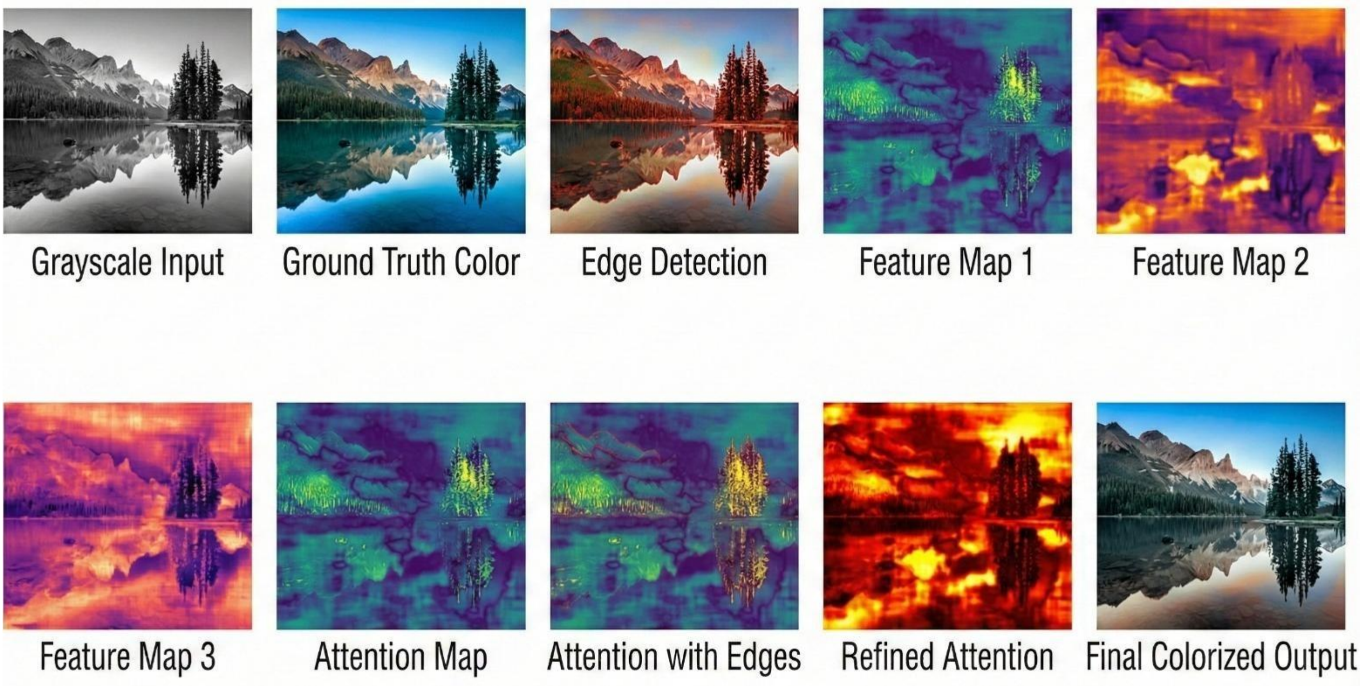

Figure 1

Figure 2

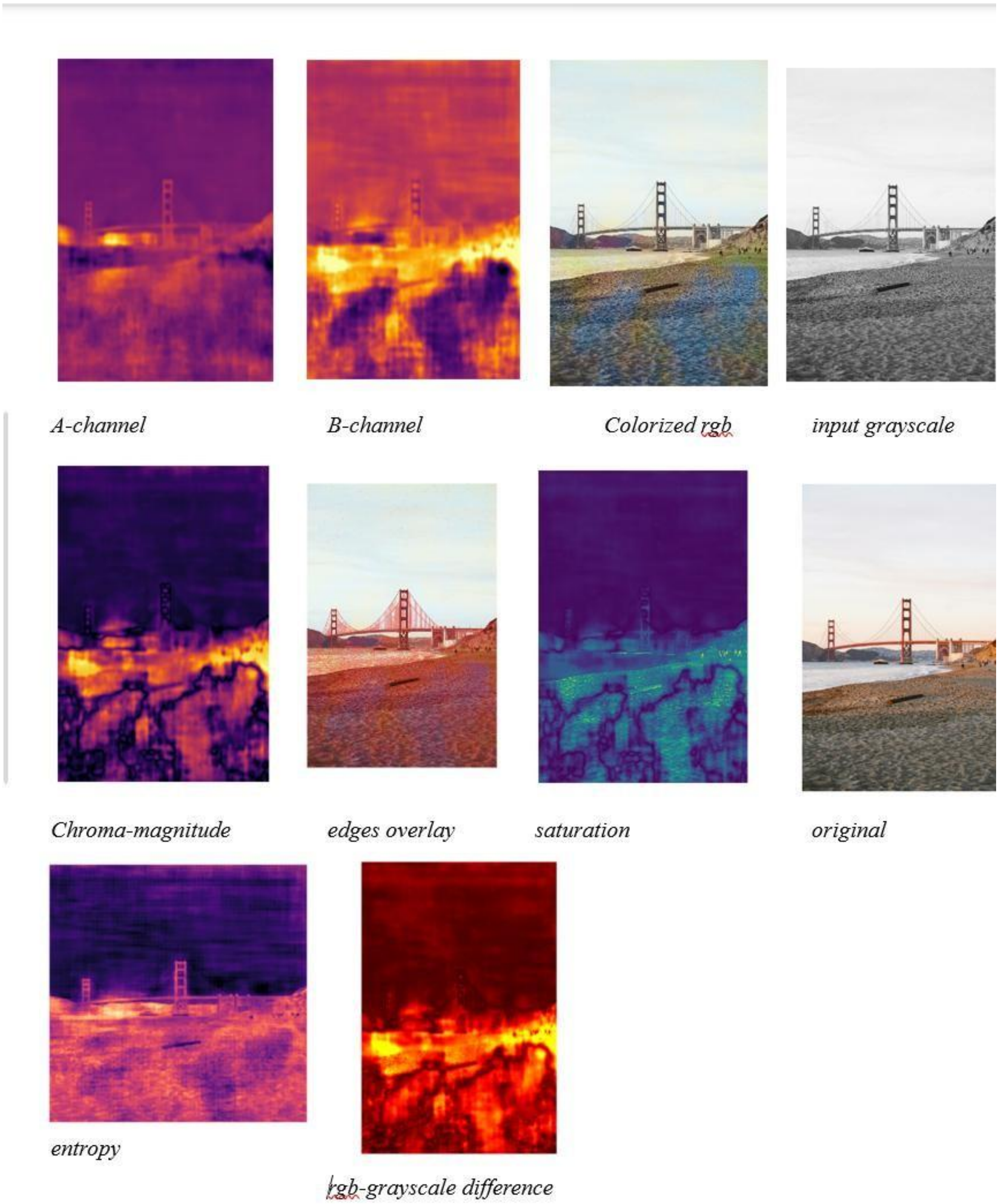

Figure 3

Lighting and Uncertainty:

The model struggles slightly in scenes with complex lighting or inconsistent lighting. In these cases, it tends to be overconfident, resulting in unrealistic color blobs on the objects. These issues could potentially be resolved with more data and training.

Example 2:

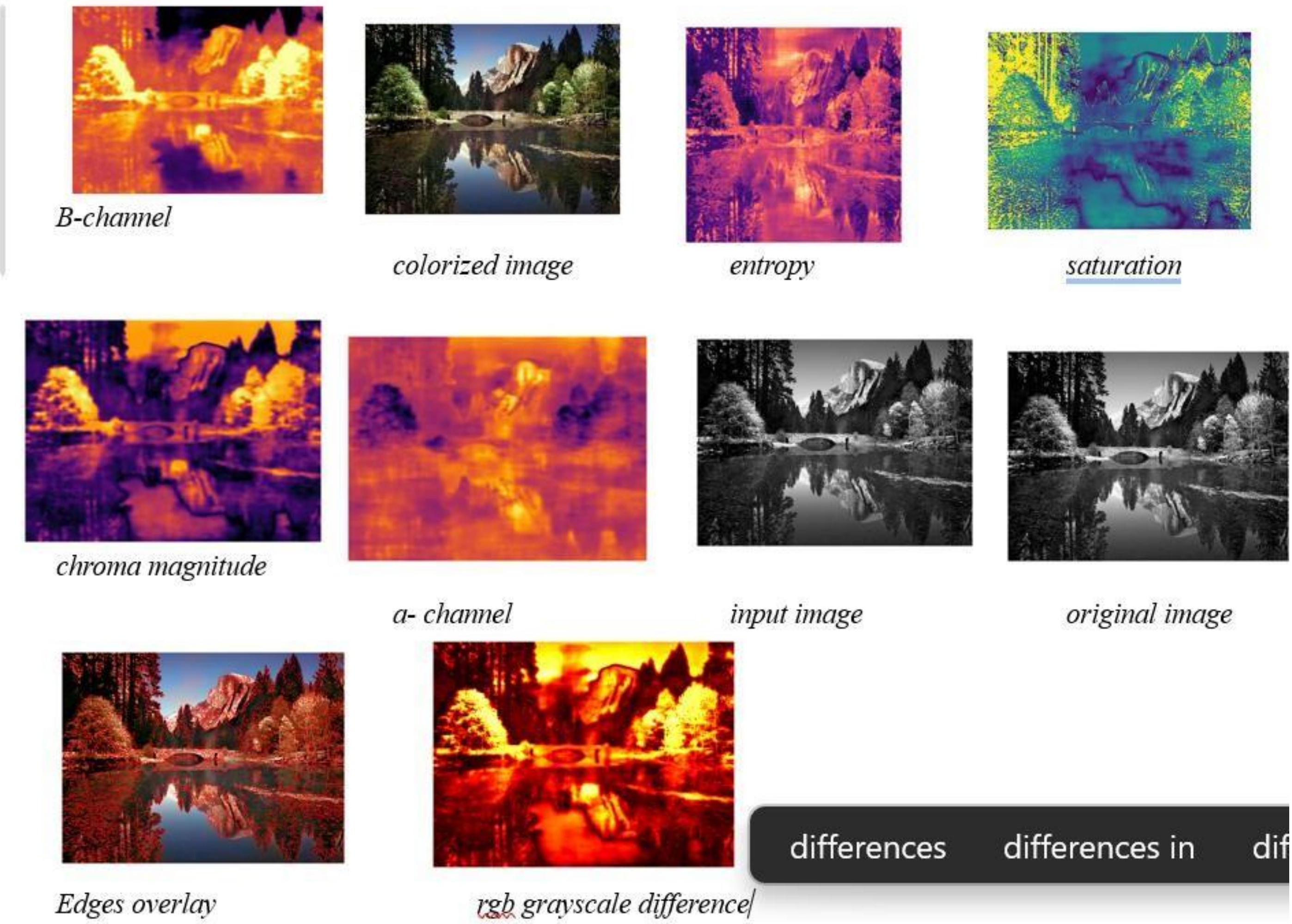

Figure 4

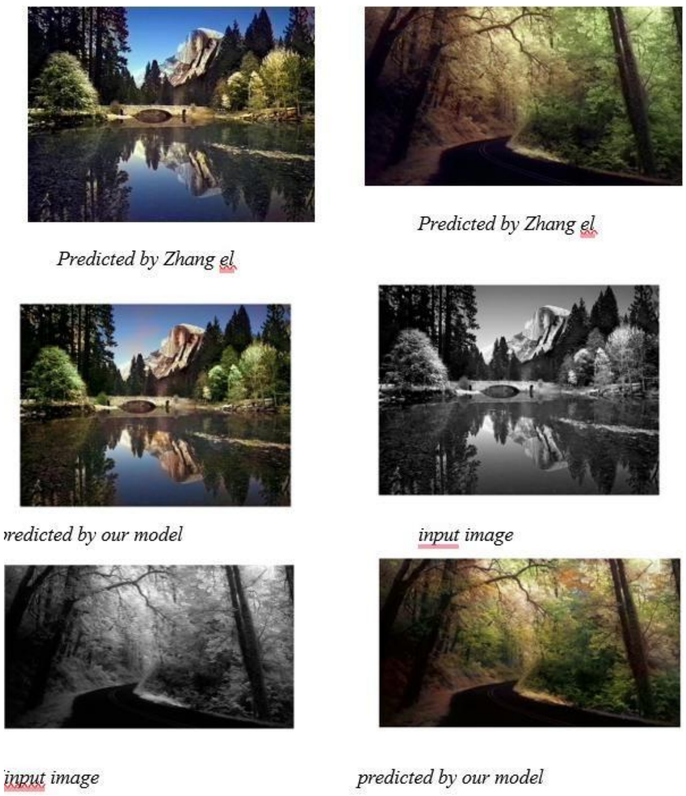

Figure 5

Potential for Improvement with More Data:

A larger, more diverse dataset (500K-1M images) could help address these issues, particularly those related to lighting and overconfident color predictions. More training could help the model better understand diverse lighting and reduce uncertainty in color generation.

2. Comparison with Other Models:

The model was compared with Zhang et al. (2016) and regression-based methods:

Comparison with Zhang et al. (2016):

Compared with Zhang et al.'s model, our model achieved similar or slightly better results. The key improvements were in:

- Color vibrancy: Our model generated more vibrant colors, with less color bleeding and a brighter overall result.
- Object boundaries: The U-Net architecture helped our model maintain better segmentation of objects, preserving more precise boundaries between different elements.

Example 3:

regression model

Original image

our model predictions

Original image

our model predictions

Figure 6

Comparison with Regression-Based Models:

Our model outperforms regression-based models in terms of:

- Color vibrancy: The output colors are brighter and more vivid, providing a richer and more dynamic colorization.
- Lighting: Our model performs better in scenes with varying lighting conditions, where the regression models tended to produce low-resolution but oversaturated outputs.

6 Conclusion

Overall, the model demonstrates strong potential in colorizing grayscale images, with good object-level colorization and minimal color bleeding. However, challenges remain with lighting and uncertainty, particularly in complex or low-light scenes. These issues could be mitigated by training on a larger, more diverse dataset (500K-1M images).

In comparison with Zhang et al. (2016), our model performs similarly or slightly better in color vibrancy and object separation. Compared with regression-based methods, our model outperforms them by producing brighter, more vivid colorizations and better handling of lighting conditions.

The model shows strong potential; moreover, further improvements are achievable with additional data and additional training.

6.1 Future Work

The model demonstrates strong potential for automatic image colorization, especially in preserving object boundaries and producing vibrant, dynamic colors. It outperforms regression-based models by creating more vivid and accurate colors, and its results are comparable to those of classification-based methods such as Zhang et al. (2016).

However, challenges remain in handling complex lighting, ambiguous objects, and rare color predictions. To address these limitations, future work could involve:

- Larger and more diverse datasets: Training on 500K to 1M images could improve the model's robustness and accuracy across different lighting conditions and object types.
- Advanced architecture: Exploring architectures like Generative Adversarial Networks (GANs) could improve the realism of the generated colors.
- Refinement with perceptual loss: Adding perceptual loss functions could help the model focus on human-perceived color quality, improving the naturalness of colorization.

Further refinement in these areas will enable the model to produce more accurate, robust colorization across a broader range of real-world images.

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